Name of Student____Form 3

MICHIRU VIEW BOYS SECONDARY SCHOOL

FORM 3 END OF TERM TWO EXAMINATION

CHEMISTRY

[100 Marks]

Wednesday, 01st June 2022

Time Allowed: 2 hours

13:30 – 15:30

Instructions

1. **This paper contains 9 printed pages. Please check.**
2. Fill in your name at the top of each page
3. Write your answers in the spaces provided
4. This paper has **two** Sections, **A** and **B**.
5. Answer **all** the **12** questions
6. Use of calculators is allowed.
7. In the table provided on this page, **tick** against the question number you have answered.
8. Hand in your paper to the invigilator when time is called to stop writing.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
Total marks scored			

Turn Over

Section A [70 marks]

Name of Student____**Form 3**

1. a) Define the term "chemical waste".

____

____ **[1 mark]**

b) What are **three** examples of wastes that are subjected to disposal?

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____ **[3 marks]**

c) Explain why is it advisable to treat chemical wastes properly before disposal?

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____ **[3 marks]**

2. a) What is a precipitate?

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____ **[1 mark]**

b**)** State the difference between a cation and anion.

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____ **[2 marks]**

c) Using ammonium hydroxide solution test, explain how the following cations can be identified from the solution.

i. Copper (II) ions

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____ **[2 marks]**

ii. Iron (II) ions

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____ **[2 marks]**

d) Explain a chemical test which shows presence of water.

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____

____ **[3 marks]**

3. Ammonia is made by Haber process according to the following chemical equation:

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

a) Name the sources of the following:

i. Hydrogen____ **[1 mark]**

ii. Nitrogen____ **[1 mark]**

b) Mention **three** reaction conditions that increase percentage of ammonia at equilibrium of the reaction.

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____ **[3 marks]**

c) Give any **two** uses of ammonia.

____

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____ **[2 marks]**

d) Briefly explain how ammonia gas can be tested.

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____ **[3 marks]**

4. a) Why is argon gas obtained before oxygen gas during separation of air into its component gases?

____

____ **[2 marks]**

b) Describe how carbon dioxide gas is removed from air.

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____ **[2 marks]**

5. a) Pure nitric acid is colourless but the nitric acid prepared in the laboratory appear yellowish in colour.

i. Briefly, explain why the nitric acid appear yellow.

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____ **[3 marks]**

ii. Give a method to turn the yellow colour back to colourless.

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____ **[1 mark]**

b) Write a balanced equation for the laboratory preparation of nitric acid.

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____ **[2 marks]**

c) Why the apparatus used in the laboratory preparation of nitric acid are all made of glass?

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____ **[2 marks]**

6. a) What is a “pure substance”

____

____ **[1 mark]**

b) Differentiate melting point from boiling point as methods of checking purity of substances.

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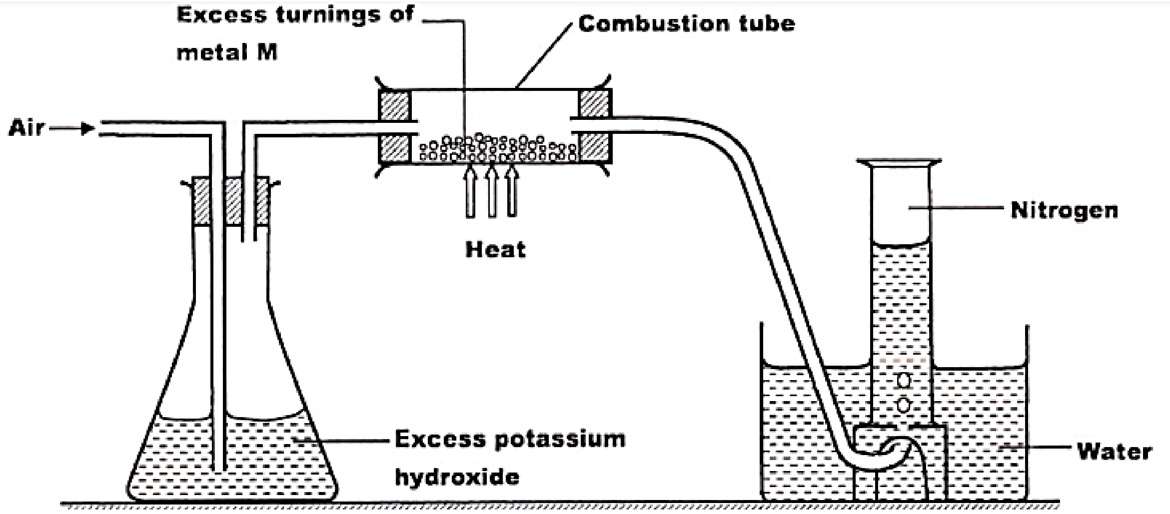
____ **[4 marks]**

d) Calculate relative flow if the distance travelled by substance is 0.013×10^3 cm and the distance moved by solvent is 0.018×10^3 cm.

[3 marks]

7. A student used the set up below to prepare a sample of nitrogen in a laboratory. Use it to answer the questions that follow.



a) State the function of potassium hydroxide in the set up.

____

____ **[1 mark]**

b) Name the metal **M** to be used in the combustion tube.

____ **[1 mark]**

c) Give a reason why the nitrogen gas obtained is not pure

____

____

____ **[2 marks]**

8. Briefly explain the following uses of nitrogen:

a) Why nitrogen is filled in an empty oil tanker.

____

____

____ **[2 marks]**

b) Why nitrogen is used for storage of semen in artificial insemination?

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____ **[2 marks]**

c) Why nitrogen is used in food processing?

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____ **[2 marks]**

9. a) Define "scientific investigation".

____

____ **[1 mark]**

b) Explain why it is important to follow the right procedure when carrying out scientific investigation.

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____ **[3 marks]**

Section B [30 marks]

Answer **all** questions in this section in the spaces provided.

10. Describe any **five** safe ways of disposing solid wastes in a laboratory.

____ **[10 marks]**

11. With the aid of a well labelled diagram, explain an experiment you would carry out in order to prepare ammonia gas in a laboratory.

[10 marks]

12. With an aid of a well labelled diagram, describe an experiment that can be conducted to prove if a certain ink sample was pure.

[10 marks]

End of Question Paper